

Laboratório IV

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1.

```
library("readxl")  
library("tidyverse")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.3      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.2  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

2.

```
fname = "../datasets/WDIEXCEL.xlsx"  
  
file.exists(fname)
```

```
[1] TRUE
```

3.

```
countries <- read_excel(fname, sheet="Country")

str(countries)
```

```
tibble [264 x 31] (S3: tbl_df/tbl/data.frame)
```

```

$ Country Code      : chr [1:264] "ABW" "AFE" "AFG" "AFW" ..
$ Short Name        : chr [1:264] "Aruba" "Africa Eastern and ...
$ Table Name        : chr [1:264] "Aruba" "Africa Eastern and ...
$ Long Name         : chr [1:264] "Aruba" "Africa Eastern and ...
$ 2-alpha code      : chr [1:264] "AW" "ZH" "AF" "ZI" ...
$ Currency Unit     : chr [1:264] "Aruban florin" NA "Afghan ...
$ Special Notes     : chr [1:264] NA "26 countries, stretchin ...
$ Region            : chr [1:264] "Latin America & Caribbean ...
$ Income Group      : chr [1:264] "High income" NA "Low incom ...
$ WB-2 code         : chr [1:264] "AW" "ZH" "AF" "ZI" ...
$ National accounts base year : chr [1:264] "2013" NA "2016" NA ...
$ National accounts reference year : num [1:264] NA NA NA NA NA 2020 NA NA ...
$ SNA price valuation : chr [1:264] "Value added at basic price ...
$ Lending category  : chr [1:264] NA NA "IDA" NA ...
$ Other groups       : chr [1:264] NA NA "HIPC" NA ...
$ System of National Accounts : chr [1:264] "Country uses the 1993 Syst ...
$ Alternative conversion factor : logi [1:264] NA NA NA NA NA NA ...
$ PPP survey year    : logi [1:264] NA NA NA NA NA NA ...
$ Balance of Payments Manual in use : chr [1:264] "BPM6" NA "BPM6" NA ...
$ External debt Reporting status : chr [1:264] NA NA "Estimate" NA ...
$ System of trade    : chr [1:264] "General trade system" NA ...
$ Government Accounting concept : chr [1:264] NA NA "Consolidated centra ...
$ IMF data dissemination standard : chr [1:264] "Enhanced General Data Dis ...
GDDS)" NA "Enhanced General Data Dissemination System (e-GDDS)" NA ...
$ Latest population census : chr [1:264] "2020 (expected)" NA "1979 ...
$ Latest household survey : chr [1:264] NA NA "Multiple Indicator ...
2023" NA ...
$ Source of most recent Income and expenditure data: chr [1:264] NA NA "Integrated household ...
$ Vital registration complete : chr [1:264] "Yes" NA NA NA ...
$ Latest agricultural census : num [1:264] NA NA 2003 NA 2023 ...
$ Latest industrial data : num [1:264] NA NA NA NA NA ...
$ Latest trade data : num [1:264] 2018 NA 2018 NA 2018 ...
$ Latest water withdrawal data : logi [1:264] NA NA NA NA NA NA ...
```

4.

```
sel_countries <- countries |> filter(`Short Name` %in% c("Afghanistan", "Brazil", "Cuba", "D  
sel_countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name`           `2-alpha code`
  <chr>          <chr>        <chr>        <chr>                <chr>
1 AFG            Afghanistan Afghanistan Islamic State of Afgh~ AF
2 BRA            Brazil      Brazil      Federative Republic o~ BR
3 CUB            Cuba        Cuba        Republic of Cuba      CU
4 DNK            Denmark     Denmark     Kingdom of Denmark    DK
5 SWZ            Eswatini     Eswatini     Kingdom of Eswatini   SZ
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

5.e 6.

```
# WDI <- read_excel(fname)
# A linha de código tradicional é muito pesada para o processamento extremamente limitado do

process_chunk <- function(chunk, pos) {
  chunk %>%
    filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
    select(`Country Code`, `Indicator Code`, `1960`:`2018`)
}

WDI <- read_csv_chunked(
  file = "../datasets/WDIEXCEL_Data.csv",
  callback = DataFrameCallback$new(process_chunk),
  chunk_size = 10000
)
```

```
-- Column specification -----
cols(
  .default = col_character()
)
i Use `spec()` for the full column specifications.
```

WDI

```
# A tibble: 265 x 61
  `Country Code` `Indicator Code` `1960`      `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <chr>      <chr> <chr> <chr> <chr> <chr>
1 AFE           SP.DYN.CBRT.IN  47,710772~ 47,73~ 47,79~ 47,83~ 47,83~ 47,83~
2 AFW           SP.DYN.CBRT.IN  47,520065~ 47,58~ 47,65~ 47,62~ 47,62~ 47,60~
3 ARB           SP.DYN.CBRT.IN  47,827211~ 47,83~ 48,21~ 47,87~ 47,42~ 46,89~
4 CSS           SP.DYN.CBRT.IN  40,861764~ 40,21~ 39,72~ 39,02~ 38,18~ 37,23~
5 CEB           SP.DYN.CBRT.IN  19,117455~ 18,10~ 17,20~ 16,99~ 16,50~ 15,93~
6 EAR           SP.DYN.CBRT.IN  44,362093~ 44,15~ 43,93~ 43,65~ 43,37~ 42,98~
7 EAS           SP.DYN.CBRT.IN  26,308286~ 24,32~ 36,38~ 40,33~ 37,50~ 36,61~
8 EAP           SP.DYN.CBRT.IN  26,640602~ 24,39~ 38,54~ 43,16~ 39,90~ 38,84~
9 TEA           SP.DYN.CBRT.IN  26,532081~ 24,29~ 38,60~ 43,25~ 39,96~ 38,88~
10 EMU          SP.DYN.CBRT.IN  18,800960~ 18,95~ 18,81~ 18,89~ 19,07~ 18,49~
# i 255 more rows
# i 53 more variables: `1966` <chr>, `1967` <chr>, `1968` <chr>, `1969` <chr>,
#   `1970` <chr>, `1971` <chr>, `1972` <chr>, `1973` <chr>, `1974` <chr>,
#   `1975` <chr>, `1976` <chr>, `1977` <chr>, `1978` <chr>, `1979` <chr>,
#   `1980` <chr>, `1981` <chr>, `1982` <chr>, `1983` <chr>, `1984` <chr>,
#   `1985` <chr>, `1986` <chr>, `1987` <chr>, `1988` <chr>, `1989` <chr>,
#   `1990` <chr>, `1991` <chr>, `1992` <chr>, `1993` <chr>, `1994` <chr>, ...
```

7. e 8.

```
WDI_long <- WDI |>
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "SP.DYN.CBRT"
  ) |>
  mutate(
    YEAR = as.integer(YEAR),
```

```

    SP.DYN.CBRT = as.numeric(str_replace(SP.DYN.CBRT, ",", "."))
  )
WDI_long

```

```
# A tibble: 15,635 x 4
```

	`Country Code` <chr>	`Indicator Code` <chr>	YEAR <int>	SP.DYN.CBRT <dbl>
1	AFE	SP.DYN.CBRT.IN	1960	47.7
2	AFE	SP.DYN.CBRT.IN	1961	47.7
3	AFE	SP.DYN.CBRT.IN	1962	47.8
4	AFE	SP.DYN.CBRT.IN	1963	47.8
5	AFE	SP.DYN.CBRT.IN	1964	47.8
6	AFE	SP.DYN.CBRT.IN	1965	47.8
7	AFE	SP.DYN.CBRT.IN	1966	47.9
8	AFE	SP.DYN.CBRT.IN	1967	47.9
9	AFE	SP.DYN.CBRT.IN	1968	47.9
10	AFE	SP.DYN.CBRT.IN	1969	47.8

```
# i 15,625 more rows
```

9. e 10.

```

sel_data <- sel_countries |>
  inner_join(WDI_long, by="Country Code") |>
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT) |> # Pode `SP.DYN.CBRT.IN`, dou s
  arrange(`Country Code`, YEAR) # Desnecessário, os dados já estavam ordenados.

str(sel_data)

```

```
tibble [295 x 4] (S3: tbl_df/tbl/data.frame)
```

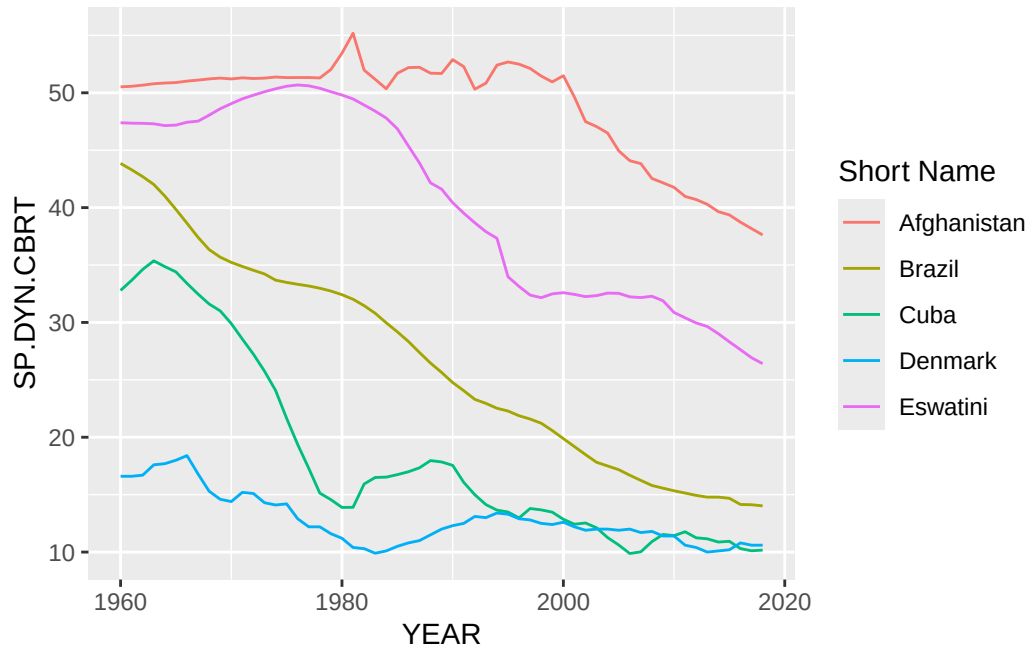
```

$ Country Code: chr [1:295] "AFG" "AFG" "AFG" "AFG" ...
$ Short Name   : chr [1:295] "Afghanistan" "Afghanistan" "Afghanistan" "Afghanistan" ...
$ YEAR         : int [1:295] 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 ...
$ SP.DYN.CBRT  : num [1:295] 50.5 50.6 50.7 50.8 50.9 ...

```

11.

```
library("ggplot2")
ggplot(sel_data, aes(x=YEAR, y=SP.DYN.CBRT, color=`Short Name`)) +
  geom_line()
```



Momento de compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-05 13:40:13 -03"
```